

Direct proof of a theorem about compact funcoids

OPG-59900

L1 - TRACTABLE

PARTIALLY SOLVED

Topology

ORIGINAL PROBLEM STATEMENT

Conjecture Let f is a T_1 -separable (the same as T_2 for symmetric transitive) compact funcoid and g is a uniform space (reflexive, symmetric, and transitive endoreloid) such that $(\text{FCD})g = f$. Then $g = \langle f \times f \rangle^* \Delta$.

The main purpose here is to find a direct proof of this conjecture. It seems that this conjecture can be derived from the well known theorem about existence of exactly one uniformity on a compact set. But that would be what I call an indirect proof, we need a direct proof instead.

The direct proof may be constructed by correcting all errors and omissions in this draft article.

Direct proof could be better because with it we would get a little more general statement like this:

Conjecture Let f be a T_1 -separable compact reflexive symmetric funcoid and g be a reloid such that

$$- (\text{FCD})g = f;$$

$$- g \circ g^{-1} \subseteq g.$$

Then $g = \langle f \times f \rangle^* \Delta$.

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

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